Wei You

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Academic Appointment

Hong Kong University of Science and Technology

Assistant Professor (2019–present), Industrial Engineering and Decision Analytics (tenure track)

Education

Columbia University

Ph.D. in Operations Research (2019)

Thesis: A Robust Queueing Network Analyzer Based on Indices of Dispersion

Committee: Ward Whitt (Advisor), Henry Lam, Martin Reiman, David Yao, Assaf Zeevi

Nanjing University

B.Sc in Mathematics (2014)

Research Interests

I study data-driven decision making at the intersection of machine learning, stochastic modeling, causal inference, and optimization, focusing on turning rigorous theory into algorithms that learn, adapt, and act reliably in digital platforms, AI deployment, revenue management, and trustworthy algorithmic decision support.

Publications & Preprints

(*Student coauthors, ♯alphabetical order, †corresponding author)

Publications

1. Xueping Gong*, **Wei You**♯† and Jiheng Zhang (2026). Efficient Transfer Learning via Causal Bounds. **Operations Research**, Forthcoming.
2. Chenguang (Allen) Wu and **Wei You**♯† (2026). Service Rate Differentiation for Homogeneous Impatient Customers. **Production and Operations Management**, Conditional Accept.
3. Zihao Wang* and **Wei You**♯† (2026). A Multi-Armed Bandit Approach to Online Resource Allocation for Service Platforms. **Naval Research Logistics**.
4. **Wei You**† (2026). Robust queueing for single-server queues with abandonment. **Queueing Systems**, 110, Article 49.
5. Qiaoqiao Wang* and **Wei You**♯† (2026). Ranking-and-Selection with Multiple Correct Answers and Non-Answerable Estimates. **Winter Simulation Conference**.
6. Xueping Gong*, **Wei You**♯† and Jiheng Zhang (2025). Minimax Optimality in Contextual Dynamic Pricing with General Valuation Models. **Operations Research**, 74(2), 879-897.

7. Chao Qin and **Wei You**$^{\sharp\dagger}$ (2025). Dual-Directed Algorithm Design for Efficient Pure Exploration. **Operations Research**, 74(2), 1104-1125.

8. Shuoguang Yang, **Wei You**, Zhe Zhang, and Ethan X. Fang (2025). Stochastic Compositional Optimization with Compositional Constraints. **INFORMS Journal on Optimization**, 8(1), 1-21.

9. **Wei You**†, Chao Qin, Zihao Wang*, Shuoguang Yang (2023). Information-Directed Selection for Top-Two Algorithms. In *The Thirty Sixth Annual Conference on Learning Theory* (**COLT 2023**), pp. 2850-2851.

10. Ward Whitt and **Wei You**$^{\sharp}$ (2022). New decomposition approximations for queueing network. **Queueing Systems**, 100(3), 365-367.

11. Ward Whitt and **Wei You**$^{\sharp}$ (2022). A Robust Queueing Network Analyzer Based on Indices of Dispersion. **Naval Research Logistics**, 69(1), 36-56.

12. Ward Whitt and **Wei You**$^{\sharp}$ (2020). Heavy-Traffic Limits for Stationary Network Flows. **Queueing Systems**, 95(1), 53-68.

13. Ward Whitt and **Wei You**$^{\sharp}$ (2019). Time-Varying Robust Queueing. **Operations Research**, 67(6), 1766-1782.

14. Ward Whitt and **Wei You**$^{\sharp}$ (2019). The Advantage of Indices of Dispersion in Queueing Approximations. **Operations Research Letters**, 47(2), 99-104.

15. Ward Whitt and **Wei You**$^{\sharp}$ (2018). Heavy-Traffic Limit of the GI/GI/1 Stationary Departure Process and its Variance Function. **Stochastic Systems**, 8(2), 143-165.

16. Ward Whitt and **Wei You**$^{\sharp}$ (2018). Using Robust Queueing to Expose the Impact of Dependence in Single-Server Queues. **Operations Research**, 66(1), 184-199.

Papers Under Review & Working Papers

1. Exponential Ergodicity and Invariant Distribution of a Degenerate Reflected SDE with Stable Noise,
   Guodong Pang and **Wei You**$^{\sharp\dagger}$, under review at **Mathematics of Operations Research**, 2026.

2. Steady-State Limit Interchange for Join-the-Shortest-Queue in the Nondegenerate Slowdown Regime,
   Wei You†, under review at **Operations Research**, 2026.

3. Cost-Aware Exploration for Structured Bandits,
   Xinyu Liu*, Chao Qin and **Wei You**$^{\sharp\dagger}$, preparing submission to **Operations Research**, 2026.

4. Pure Exploration via Frank–Wolfe Self-Play,
   Xinyu Liu*, Chao Qin and **Wei You**$^{\sharp\dagger}$, preparing submission to **Operations Research**, 2026.

5. Principal Fairness with Optimal Transport for Heterogeneous Population,
   Huichen Zhu, **Wei You** and Annie Qu, under review at **Journal of the American Statistical Association**, 2026.

Teaching Experience

Hong Kong University of Science and Technology, Instructor

IEDA3250 Stochastic Models (2026–present)

IEDA5270 Engineering Statistics and Data Analytics (2019–2025)

IEDA2540 Statistics for Engineers (2020–2025)

EEMT5500 Applied Probability, Statistics and Data Analytics (2022)

Columbia University, Teaching Assistant

Applied Statistical Models (2017)

Simulation (2017, 2015)

Stochastic Models (2014)

Conference & Invited Talks

(*Presented by co-authors)

Conferences

"Minimax Optimality in Contextual Dynamic Pricing with General Valuation Models"
2026 POMS International Conference in China, Xi'an*

"Pure Exploration via Frank–Wolfe Self-Play"
POMS-HK 2026, Shenzhen*
NeurIPS 2025 ML×OR Workshop, San Diego, CA*
APORS Youth Forum 2025, Hong Kong*

"Transfer Learning with Partially Observable Offline Data via Causal Bounds"
POMS-HK 2026, Shenzhen*
EcoSta 2025, Tokyo, Japan

"Cost-Aware Exploration for Structured Bandits"
2025 INFORMS Applied Probability Conference, Atlanta, GA
2025 INFORMS International Meeting, Singapore

"Dual-Directed Algorithm Design for Efficient Pure Exploration"
2024 INFORMS Annual Meeting, Seattle, WA
IASC-ARS Interim Conference 2024, Taipei City
JCSDS 2023, Kunming, Yunnan

"Information-Directed Selection for Pure Exploration"
2023 INFORMS Annual Meeting, Phoenix, AZ

"Information-Directed Selection for Top-Two Algorithms"
Conference on Learning Theory (COLT) 2023, Bangalore, India
2023 INFORMS APS Meeting, Nancy, France
POMS-HK 2023, Hong Kong

"Approximation for Single-Server Queues with Abandonment via Robust Queueing"
2024 INFORMS MSOM Conference, Minneapolis, Minnesota
2021 INFORMS Annual Meeting, Anaheim, CA

"Online Resource Allocation for Service Platforms"
Virtual 2020 INFORMS Annual Meeting

"Heavy-Traffic Limit for Stationary Network Flows"
2019 INFORMS APS Meeting, Brisbane, Australia

"Time-Varying Robust Queueing for Quantile Approxiamtion"
POMS-HK 2020, Hong Kong
2019 INFORMS APS Meeting, Brisbane, Australia

"A Robust Queueing Network Analyzer Based on Indices of Dispersion"
2018 INFORMS Annual Meeting, Phoenix, AZ

"A Robust Queueing Network Analyzer for a Series of Single-Server Queues"
2017 INFORMS Annual Meeting, Houston, TX

"Time-Varying Robust Queueing"
2017 INFORMS APS Meeting, Evanston, IL

"Using Robust Queueing To Expose The Impact of Dependence In Single Server Queues"
2016 INFORMS Annual Meeting, Nashville, TN

Invited Talks

Miami Herbert Business School, University of Miami, 2025

Antai College of Economics & Management, Shanghai Jiao Tong University, 2025

Institute of Statistics and Big Data, Renmin University of China, 2024

School of Statistics and Data Science, Nankai University, 2024

School of Management, Fudan University, 2024

School of Digital Economics, Shanghai University of Finance and Economics, 2023

School of Management and Engineering, Nanjing University, 2017, 2023

Institute of Applied Mathematics, Chinese Academy of Sciences, 2019

Institute of Mathematical Science, ShanghaiTech University, 2019

Industrial Engineering and Decision Analytics, HKUST, 2019

Industrial and System Engineering, Virginia Tech, 2019

System and Industrial Engineering, University of Arizona, 2019

Business School, University of Miami, 2019

School of Management, University of Science and Technology of China, 2019

Institute for Data and Decision Analytics, Chinese University of Hong Kong, Shen Zhen, 2019

System Engineering and Engineering Management, Chinese University of Hong Kong, 2019

Grants

Hong Kong RGC–Theme-based Research Scheme (2025–2029). Project No. T32-615/24R, Co-PI. HK$61,191,000.

Hong Kong RGC-GRF (2024–2026). Project No. 16212823, PI, HK$651,464.

WeBank (2021–2024). Project No. WEB19EG01-M, Co-PI, HK$684,000.

Hong Kong RGC-ECS (2021–2023). Project No. 26212320, PI, HK$529,972.

HKUST Start-up Fund. PI, HK$1,000,000.

Students

Xinyu Liu, Ph.D. (2024–present)

Yueying Hu, Ph.D. (2024–present)

Qiaoqiao Wang, Ph.D. (2023–present)

Ke Lu, Mphil. (2023–present)

Zihao Wang, Ph.D. (2019–2023)
— Thesis: "Online Resource Allocation for Service Platforms and Bandit Experiments"
— First placement: Equity Derivatives Quant, Bank of America.

Yingxian Wu, Mphil. (2020–2022), co-advised with Prof. Sisi Jian
— Thesis: "Mix Allocation of Two Types of Passengers and Drivers"
— First placement: Investment Risk Management Assistant, E Fund Management Co., LTD.

Academic Service

Reviewer: Management Science, Operations Research, Manufacturing & Service Operations Management, Queueing Systems, Stochastic Systems, Performance Evaluation, Production and Operations Management, Naval Research Logistics, Probability in the Engineering and Informational Sciences, Operations Research Letters, Israel Science Foundation (ISF)

Program committee: Winter Simulation Conference (2024, 2025, 2026), Algorithmic Learning Theory (2025, 2026, 2027), NeurIPS 2026 Workshop MLxOR (2026)

Judge: POMS-HK Best Student Paper Award (2019, 2023), 2026 APS Student Paper Competition (2026)

Session Chair: 2017 INFORMS APS Meeting, POMS-HK (2023, 2025), 2025 INFORMS Applied Probability Conference, 2025 INFORMS International Meeting, 2027 ASCI Conference